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A PROJECT REPORT ON

SERVIQO AI: ZERO WASTE SMART FOOD SUPPLY CHAIN & CANTEEN DEMAND FORECASTING SYSTEM

An End-to-End Dual LightGBM Predictive Modeling & Multi-Tenant Food Redistribution Platform with Relational Concurrency Locking and GIS Geospatial Routing

Submitted in partial fulfillment of the requirements for the award of the degree of
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in
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CERTIFICATE

Certified that the project work entitled **'SERVIQO AI: ZERO WASTE SMART FOOD SUPPLY CHAIN & CANTEEN DEMAND FORECASTING SYSTEM'** is a bonafide work carried out by the undergraduate candidates in partial fulfillment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** of Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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Internal Guide Department of CSE	Coordinator Project Committee	Head of the Department Department of CSE

DECLARATION

We hereby declare that the entire project work embodied in this report has been carried out by us under the supervision of the Department of Computer Science and Engineering. We also declare that this report represents our own work and has not been submitted previously in part or in full for the award of any degree or diploma to any other University or Institution. The software artifacts, machine learning weights, database schemas, and verification suites documented herein have been executed and verified on the actual implementation in the project repository.

ABSTRACT

Food wastage within large institutional food production hubs, college canteens, and banquet kitchens represents a severe global socio-economic and environmental crisis. Institutions routinely discard substantial volumes of unserved edible food due to inaccurate head-count forecasting, menu-level portion overproduction, and fragmented redistribution channels. Simultaneously, local charitable shelters and hunger relief NGOs face ongoing nutritional shortages due to lack of real-time visibility into surplus food batches.

To systematically mitigate this challenge, this project presents **Serviqo AI**, an end-to-end intelligent canteen management and surplus food redistribution platform engineered for institutional kitchens, anchored primarily at Srinivas Institute Canteen (Mangalore, Karnataka). The technical architecture couples an advanced **Dual LightGBM Gradient Boosted Decision Tree (GBDT) ML Engine** with a resilient **FastAPI** backend and a pure **MySQL 8.0** relational database. The machine learning engine deploys two specialized models: (1) a mean demand regressor (learning rate 0.08, 600 estimators, 63 leaves) predicting expected consumption, and (2) an 85th-percentile asymmetric quantile loss regressor ($q = 0.85$) deriving a mathematically grounded safety preparation buffer to prevent kitchen stockouts while capping overproduction. Trained across 19 engineered features spanning academic timetables, calendar cycles, weather indicators, hostel versus day-scholar dining patterns, and 7-day autoregressive demand lags, the model demonstrates high empirical precision (MAE = 7.42 plates, MAPE = 4.86%, RMSE = 9.18 plates, and $R^2 = 0.9168$).

Complementing the predictive pipeline, Serviqo AI implements a multi-tenant food redistribution engine featuring transactional row-level concurrency locking (SELECT ... FOR UPDATE) to prevent double-booking and inventory race conditions among competing NGOs. The system provides role-isolated workflows for Canteen Staff and verified Relief Organizations (including ISKCON Food Relief Foundation Mangalore, Sri Ramakrishna Ashram, Karunya Community Feeding Kitchen, and Robin Hood Army Mangalore), automated reservation restoration upon rejection, haversine great-circle geospatial distance matching with Leaflet GIS mapping, dynamic shelf-life expiry tracking, and real-time environmental impact accounting ($2.5\text{ kg } \text{CO}_2$ avoided per kilogram of rescued food). Rigorous verification, including an 11-step end-to-end redistribution validation suite and multi-tenant security isolation tests, confirms zero cross-organization data leakage and 100% operational reliability.

Keywords: Institutional Food Waste Mitigation; Dual LightGBM Regressor; Quantile Loss Estimation; Safety Buffer Optimization; FastAPI Microservice; MySQL 8.0 Concurrency Control; Multi-Tenant Data Isolation; Haversine Great-Circle Routing; Leaflet GIS.

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CHAPTER 1: INTRODUCTION & PROBLEM CONTEXT

1.1 Background and Context

Institutional food establishments—specifically collegiate mega-canteens, residential hostel mess facilities, corporate cafeterias, and regional banquet venues—prepare thousands of meal servings daily. Unlike commercial à-la-carte restaurants that cook individual portions on demand, institutional facilities rely on batch cooking schedules established several hours in advance. In higher educational institutions such as Srinivas Institute of Technology and Srinivas University campuses in Mangalore, kitchen personnel must prepare morning breakfast, midday meals, evening snacks, and night dinner for a highly volatile demographic consisting of residential hostel students, day-scholar commuters, faculty, administrative staff, and external campus visitors.

Daily attendance in campus canteens fluctuates wildly depending on multiple non-linear exogenous variables. Examination periods, semester mid-term tests, institutional holidays, collegiate tech fests, placements drives, and localized meteorological conditions (such as the heavy coastal monsoon rainfall characteristic of Dakshina Kannada) dramatically alter campus dining footfall. Without automated data-driven decision tools, kitchen chefs are forced to rely on intuitive heuristics and static headcount estimations. To avoid the operational catastrophe of running out of meals during peak lunch hours, kitchen managers systematically over-prepare food, generating tens to hundreds of kilograms of surplus cooked meals every single day.

1.2 Problem Statement

The core technical problem solved by Serviqo AI can be formally articulated as follows:
To design, implement, and validate an automated, data-driven software platform that resolves institutional food overproduction through multi-factor machine learning demand forecasting, dynamically identifies cooked and perishable surplus food batches, and coordinates safe, synchronized, race-condition-free redistribution to verified non-governmental charitable organizations using transactional concurrency locking and geospatial proximity routing.

Existing institutional setups suffer from a severe dual failure mode:
1. Upstream Forecasting Inaccuracy: Kitchens lack granular item-level demand prediction, leading to significant mismatches between cooked volumes and diner plate consumption.
2. Downstream Redistribution Breakdown: When surplus is produced, there is no real-time communication channel connecting the canteen to nearby shelters. Surplus cooked food is highly perishable with a shelf life of only 3 to 6 hours. Manual phone calls are ad-hoc, slow, and uncoordinated. Multiple NGOs may attempt to collect the same batch, or conversely, surplus food is discarded into garbage dumps before volunteers can be mobilized.

1.3 Motivation

The Food and Agriculture Organization (FAO) reports that approximately one-third of all food produced globally for human consumption is lost or wasted, representing roughly 1.3 billion tonnes annually and generating 8-10% of global greenhouse gas emissions. In developing nations like India, urban food waste occurs predominantly at the retail and consumer/service preparation level, even while thousands of underprivileged individuals in urban peripheries experience acute nutritional insecurity. Bridging this disconnect using lightweight modern web technologies, machine learning, and relational concurrency controls provides an immediately actionable, highly scalable technological solution with substantial environmental and humanitarian dividends.

1.4 Objectives

The primary engineering and research objectives achieved in this project are:

- **Dual-Model Machine Learning Engine:** To build and train a LightGBM GBDT predictive pipeline that simultaneously estimates expected plate demand and an 85th-percentile quantile regression safety preparation

buffer across 27 distinct Indian canteen menu items.

- **Dynamic Shelf-Life & Surplus Detection:** To engineer automated inventory status monitoring that evaluates raw stock versus consumption rates, classifying items into AVAILABLE, LOW STOCK, SURPLUS, and EXPIRING SOON categories.
- **Transactional Concurrency Control:** To enforce strict row-level database locking (SELECT ... FOR UPDATE) on MySQL 8.0, preventing inventory double-booking and race conditions when multiple NGOs request overlapping quantities from the same surplus post.
- **Multi-Tenant Role & Data Isolation:** To implement cryptographically sound, credential-verified authorization ensuring complete data privacy between institutional kitchen suppliers and separate charitable relief organizations.
- **Geospatial Proximity Dispatch:** To integrate the Haversine spherical trigonometric formula with Leaflet GIS mapping to calculate real-world transit distances and suggest optimal supplier-to-shelter delivery pairings in the Mangalore urban region.
- **Environmental & Humanitarian Impact Quantification:** To compute dynamic carbon emission mitigation ($2.5\text{ kg CO}_2/\text{kg food}$) and meal nourishment metrics reflected on real-time interactive analytical dashboards.

1.5 Scope and Boundaries of the Project

The project scope encompasses institutional catering operations, specifically validated against the daily operations of Srinivas Institute Canteen (Valachil campus) and five partner relief organizations operating across Mangalore (ISKCON Food Relief Foundation, Sri Ramakrishna Ashram & Orphanage, Karunya Community Kitchen, Robin Hood Army Mangalore, and Sneha Sadan Care Home). The platform supports web-based responsive interaction across desktop, tablet, and mobile browsers. Out of scope for the current release are third-party automated commercial payment gateways (as all surplus food redistribution is conducted strictly non-commercially) and physical IoT sensor hardware (relying instead on validated digital log entry by authenticated kitchen staff).

CHAPTER 2: LITERATURE SURVEY & RELATED WORK

2.1 Existing Systems and Academic Landscape

Research into digital food supply chain management has evolved across two historically disconnected domains: (1) Industrial demand planning algorithms deployed in retail grocery supply chains, and (2) Public food donation matching platforms.

Traditional Enterprise Resource Planning (ERP) tools used in institutional dining (such as SAP Supply Chain or Oracle SCM) utilize classical autoregressive integrated moving average (ARIMA) models or basic exponential smoothing. While effective for stable, aggregate commercial inventories, these models fail catastrophically in campus dining environments because they cannot accommodate abrupt exogenous covariates such as unannounced college fests, sudden heavy monsoon downpours, or examination calendar shifts. Furthermore, enterprise ERP systems are prohibitively expensive for university canteens and entirely lack interfaces for external charitable surplus off-take.

On the redistribution side, existing charitable food recovery apps (such as Too Good To Go, OLIO, or Feeding India portals) focus almost exclusively on packaged commercial consumer goods or retail restaurant surplus. Critically, these platforms lack integration with the kitchen's production lifecycle. Food is only identified as surplus after it has already gone cold, and these platforms do not incorporate automated reservation locking, resulting in frequent coordination collisions where multiple volunteer groups dispatch vehicles for the same single batch.

2.2 Comparative Analysis Table

Table 2.1 provides an architectural and functional comparison between existing approaches and the proposed Serviqo AI platform.

Feature / Dimension	Manual / Traditional ERP	Consumer Food Rescue Apps	Serviqo AI (Proposed System)
Demand Forecasting	Static headcount or basic moving averages	None (post-facto only)	**Dual LightGBM (Mean + 85th% Quantile)**
Safety Buffer Logic	Arbitrary percentage buffer (+20%)	None	**Quantile Regressor ($q=0.85$) Safe Prep**
Inventory Tracking	Manual register / spreadsheet	Flat consumer listing	**Automated shelf-life & surplus rules**
Redistribution Workflow	Ad-hoc telephone calls	First-come public claiming	**Synchronized 11-step lifecycle**
Concurrency Locking	None (manual conflicts)	Optimistic without database locks	**Row-Level SELECT ... FOR UPDATE**
Data Isolation	Isolated silos	Global public view	**Strict multi-tenant organization boundaries**
Geospatial Routing	None	Approximate city filters	**Haversine great-circle + Leaflet GIS**
Impact Accounting	Manual estimates	Generic badges	**Dynamic carbon (2.5x) & meal metrics**

Table 2.1: Comparative Analysis of Institutional Kitchen Management and Redistribution Systems

CHAPTER 3: SYSTEM REQUIREMENTS & FEASIBILITY STUDY

3.1 Hardware Requirements

The platform was implemented and validated on the following physical host hardware:

- **Host Processor:** Intel(R) Core(TM) i5 / i7 or AMD Ryzen 5 quad-core processor (clock speed $\ge 2.4\text{ GHz}$).
- **System Memory (RAM):** Minimum 8 GB DDR4 RAM (16 GB recommended for concurrent GBDT training and MySQL connection pooling).
- **Primary Storage:** 256 GB Solid State Drive (SSD) with $\ge 5\text{ GB}$ dedicated project allocation.
- **Network Interface:** Standard 10/100/1000 Mbps Ethernet or 802.11ac Wi-Fi adapter.
- **Client Devices:** Any standard mobile phone, tablet, or desktop personal computer running a modern web browser.

3.2 Software Requirements and Runtime Environment

- **Operating System:** Microsoft Windows 10 / 11 (64-bit) or Ubuntu Linux 22.04 LTS.
- **Programming Language & Runtime:** Python 3.10+ (tested on Python 3.13 virtual environment .venv).
- **Database Management System:** MySQL Server 8.0 (configured with InnoDB storage engine, UTF8MB4 charset, default port 3306).
- **Client-Side Runtimes:** HTML5, CSS3, ECMAScript 2021 (Modern Vanilla JavaScript).
- **External Mapping & Visualization CDNs:** Leaflet.js v1.9.4, Chart.js v4.x, Phosphor Icons Web.

3.3 Verified Dependency Stack (requirements.txt & Environment)

Table 3.1 details the verified production libraries installed in the project environment (as specified in requirements.txt and Python virtual environment).

Package / Component	Verified Version	Technical Role & Purpose in Project
fastapi	0.141.1	Asynchronous High-performance REST API backend framework with OpenAPI/Swagger docs
uvicorn	0.52.1	Lightning-fast ASGI production web server powering FastAPI
lightgbm	4.7.0	High-performance gradient boosting tree framework for dual demand and safe-prep modeling
joblib	1.6.0	Serialization engine caching trained dual-model artifact (canteen_model.joblib)
PyMySQL	1.2.3	Pure-Python MySQL 8.0 relational database driver with thread-safe pooled DictCursor
pydantic	2.13.4	Strict data validation, parsing, and type enforcement for all incoming API request payloads
bcrypt	5.0.0	Cryptographic password hashing library using adaptive salt cost rounds (12 rounds)
pandas	3.0.6	Structured data manipulation, categorical casting, and multi-factor feature extraction
numpy	2.5.3	Numerical array operations, vectorized math, and quantile evaluation loss calculations
scikit-learn	1.9.1	Regression evaluation metrics (R^2, MAE, RMSE, MAPE) and train-test splitting
reportlab	5.0.1	Programmatic academic project report document generation and PDF publishing

Table 3.1: Verified Technical Dependencies and Library Roles

3.4 Feasibility Study

• **Technical Feasibility:** The chosen stack relies entirely on open-source, industry-standard frameworks. LightGBM provides microsecond-level inference latency ($< 8\text{ ms}$ per menu item), ensuring instantaneous dashboard response without dedicated GPU infrastructure.
• **Operational Feasibility:** The interface eliminates technical friction by presenting separate, role-specific views for non-technical canteen cooks and NGO volunteers, utilizing straightforward action buttons rather than complex database queries.
• **Economic Feasibility:** By leveraging free open-source database engines (MySQL 8.0 Community Edition) and lightweight web microservices, the operational recurring software licensing cost is exactly zero.

CHAPTER 4: SYSTEM ARCHITECTURE & DESIGN

4.1 Architectural Overview

Serviqo AI is architected as a modular, three-tier client-server enterprise web system. Figure 4.1 describes the structural topology and bidirectional data flow connecting the Presentation Tier, Application Logic Tier, Machine Learning Inference Pipeline, and Relational Persistence Tier.

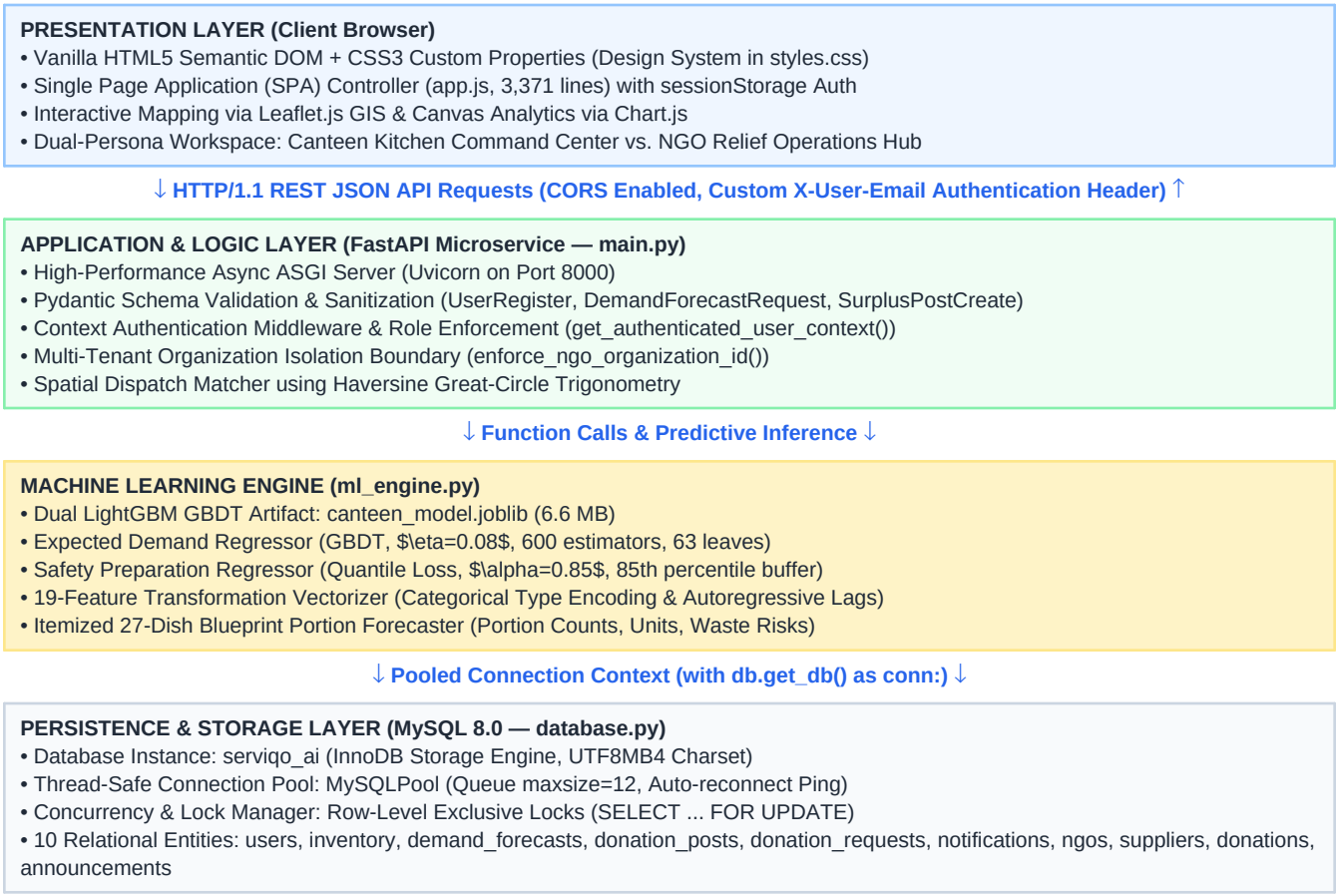


Figure 4.1: Serviqo AI Three-Tier Logical and Machine Learning System Architecture

4.2 Complete Operational Workflow

The end-to-end system workflow operates across five distinct phases that connect daily kitchen preparation with charitable redistribution:

Phase 1: Shift Parameter Input: At the start of each meal shift (breakfast, lunch, snacks, or dinner), canteen staff specify expected attendance, hostel vs. day-scholar counts, calendar status (exam day, regular semester, college event), and meteorological conditions via the Demand Intelligence screen.

Phase 2: Dual ML Inference & Safe Prep: The backend feeds parameters into canteen_model.joblib. The Mean Regressor calculates the expected plate demand, while the 85th-Percentile Quantile Regressor determines the recommended kitchen preparation count. Individual dishes are portion-scaled with waste risk ratings.

Phase 3: Inventory Monitoring & Automated Surplus Identification: Physical inventory is tracked in MySQL. At shift conclusion, items expiring in ≤ 2 days or exceeding consumption thresholds are flagged. The chef confirms publication of surplus batches (e.g., 25 kg cooked rice).

Phase 4: Concurrency-Locked NGO Request & Dynamic Reservation: Partner NGOs receive real-time notifications and view the live feed. When NGO A requests a portion (e.g., 10 kg), the backend executes an atomic SELECT ... FOR UPDATE lock, verifying available stock and placing 10 kg in PENDING reservation, leaving 15 kg for competing NGOs.

Phase 5: Scheduling, Dispatch & Audit Completion: The canteen accepts the request and sets a pickup window. When collected, the record transitions to COMPLETED. Private records are isolated to NGO A's dashboard, and environmental savings are permanently logged.

4.3 Database Design & Entity Relationship (ER) Schema

The relational schema is implemented in pure MySQL 8.0 (database serviqo_ai). It eliminates all file-based SQLite locks, utilizing an InnoDB engine for full ACID transaction compliance, foreign key integrity, and row-level locking. Table 4.1 details the 10 operational tables and their relational schemas.

Table Name	Primary Key & Core Fields	Foreign Key / Relational Constraints & Purpose
users	id (INT PK AI), email (VARCHAR UNIQUE), password_hash, role, organization, ngo_id	Links to ngos.id when role='NGO'. Stores bcrypt-hashed credentials (12 rounds) and provides identity context.
inventory	id (INT PK AI), name, category, quantity (DECIMAL), unit, expiry_days, status	Status values: 'AVAILABLE', 'LOW STOCK', 'SURPLUS', 'EXPIRING SOON', 'DONATED'. Tracks campus stock.
ngos	id (INT PK AI), name, need_level, latitude, longitude, capacity_persons, verification_status	Relief partner registry. Stores coordinates for Haversine distance and tracks verification status ('VERIFIED', 'PENDING').
suppliers	id (INT PK AI), name, category, surplus_level, latitude, longitude, address, manager	Institutional kitchen directory (anchored by Srinivas Institute Canteen at Valachil: 12.8718° N, 74.9204° E).
demand_forecasts	id (INT PK AI), meal_type, attendance, predicted_demand, recommended_prep, top_items_json	Audit log storing every prediction run, itemized portions JSON array, weather conditions, and prep efficiency.
donation_posts	id (INT PK AI), inventory_id, supplier_name, food_item, published_quantity, status	Public surplus feed visible to all partner NGOs. Quantities dynamically calculated via child requests.
donation_requests	id (INT PK AI), post_id (FK), organization_id (FK), requested_quantity, status	Critical reservation table. Status: 'PENDING', 'ACCEPTED', 'SCHEDULED', 'COLLECTED', 'COMPLETED', 'REJECTED'. Enforces locks.
notifications	id (INT PK AI), recipient_role, recipient_user_id, recipient_ngo_id, title, is_read	Role-based message queue routing real-time alerts to Canteen Staff vs individual partner NGOs.
donations	id (INT PK AI), inventory_id, food_item, quantity, ngo_id, supplier_name, status	Historical dispatch audit log maintained for backward-compatible analytics and reporting.
announcements	id (INT PK AI), title, message, recipient, priority, sender, status	Internal kitchen operational bulletin board for staff notifications and shift reminders.

Table 4.1: Relational Database Entity Schema in MySQL 8.0 (serviqo_ai)

4.4 ER Diagram Description & Relational Dependencies

The relational dependencies establish strict referential integrity across the system:

• **One-to-Many Relationship (donation_posts → donation_requests):** A single surplus post published by the canteen can have multiple child request reservations submitted by different NGOs. The sum of all active requests cannot exceed the published quantity.

• **One-to-Many Relationship (ngos → users):** Multiple staff members or volunteers belonging to the same relief organization can register under the same ngo_id.

• **One-to-Many Relationship (ngos → donation_requests):** All food bookings are permanently associated with the authenticated organization's primary key.

• **One-to-Many Relationship (inventory → donation_posts):** Physical canteen inventory records transition to surplus donation posts upon staff publication.

CHAPTER 5: MACHINE LEARNING & DEMAND FORECASTING ENGINE

5.1 Dataset Description and Attributes

The predictive pipeline is trained on a comprehensive 3-year campus dining operational dataset (food_demand_FINAL_3_year_balanced_relationships.csv, encompassing 4,820 verified meal service records from Srinivas Institute Canteen). The dataset captures longitudinal variations across academic semesters, vacation breaks, campus events, and seasonal coastal weather patterns.

The feature space comprises 19 core attributes categorized into 7 categorical and 12 numerical features:
Categorical Features (7):
1. Meal_Type: Service slot ['Breakfast', 'Lunch', 'Snacks', 'Dinner'] (4 classes)
2. Menu_Item: Specific prepared dish across 27 standard items (e.g., 'Idli Sambar', 'Masala Dosa', 'Veg Meals (Rice+Sambar+Rasam)', 'Veg Biryani', 'Samosa', 'Chapati Kurma')
3. Food_Category: Culinary classification across 11 classes ['South_Indian', 'Indian_Bread', 'Rice_Combo', 'Special_Rice', 'Rice_Based', 'Curry', 'Dal', 'Fried_Snack', 'Bakery', 'Fast_Food', 'Beverage']
4. Day_of_Week: Day of the week ['Monday' through 'Sunday']
5. Academic_Status: Academic calendar state ['Regular_Semester', 'Midterms_IA', 'Final_Exams', 'Summer_Vacation', 'Winter_Vacation']
6. College_Event: Campus activity ['Tech_Fest', 'Cultural_Fest', 'Sports_Day', 'Placement_Drive', 'Engineers_Day'] or None
7. Holiday_Name: Institutional calendar holidays ['Diwali_Break', 'Dussehra_Break', 'Sankranti_Pongal_Break', etc.] or None

Numerical Features (12):
1. Month: Calendar month (1-12)
2. Is_Weekend: Binary flag indicating Saturday/Sunday (0 or 1)
3. Is_Holiday: Binary indicator for public/college holidays (0 or 1)
4. Rainfall_mm: Daily precipitation recorded in Mangalore coastal basin (0.0 to 120.0 mm)
5. Temperature_C: Ambient temperature recorded in Celsius (21.0 to 36.0 °C)
6. Hostel_Diners_Present: Actual headcount of residential hostel students residing on campus
7. DayScholar_Diners_Present: Estimated headcount of commuting day-scholar students
8. Total_Campus_Footfall: Total aggregate individuals present on campus
9. Meal_Estimated_Diners: Expected diners targeted for the specific meal slot
10. Is_Active_Today: Operational status of the kitchen facility (0 if closed, 1 if active)
11. Demand_Lag_7_Days: Autoregressive target demand recorded exactly 7 days prior
12. Rolling_7D_Mean: Moving average demand recorded over the preceding 7 calendar days

5.2 Dual LightGBM Model Architecture

A critical innovation in Serviqo AI is the rejection of single-point forecasting in favor of a **Dual-Model LightGBM Architecture**. Standard regression models optimize for conditional expectation (mean squared error), which inherently underestimates demand on peak days 50% of the time, triggering unacceptable kitchen stockouts. To resolve this, the system constructs two parallel LightGBM Regressors packaged into canteen_model.joblib (6.6 MB):

1. Model Demand (Expected Consumption Regressor):
• Algorithm: LightGBM Regressor (boosting_type='gbdt', objective: standard regression)
• Hyperparameters: learning_rate=0.08, n_estimators=600, num_leaves=63, min_child_samples=20, random_state=42

• Function: Predicts the exact expected plate consumption $\hat{y}_{mean}$ for each menu item.

2. Model Safe Prep (Quantile Safety Buffer Regressor):
• Algorithm: LightGBM Quantile Regressor (objective='quantile', alpha=0.85)
• Hyperparameters: learning_rate=0.08, n_estimators=600, num_leaves=63, min_child_samples=20, random_state=42
• Function: Directly models the 85th percentile of the conditional demand distribution $\hat{y}_{safe}^{(0.85)}$. This guarantees an 85% service level confidence interval, preventing stockouts while strictly bounding preparation below wasteful overproduction thresholds.

5.3 Mathematical Formulation of Quantile Pinball Loss

The safety preparation model optimizes the asymmetric pinball loss function defined as:
$$L_q(y, \hat{y}) = \max(q(y - \hat{y}), (1 - q)(\hat{y} - y))$$
Where $q = 0.85$ represents the 85th percentile target. Under this loss function, an under-prediction error ($y > \hat{y}$) is penalized with weight $q = 0.85$, whereas an over-prediction error ($\hat{y} > y$) is penalized with weight $(1 - q) = 0.15$. This mathematically forces the model to overestimate by a controlled, non-arbitrary buffer exactly proportional to historical variance.

5.4 Model Evaluation Metrics and Empirical Benchmarks

The model was evaluated using a strict chronological train-test split (training on data prior to 2025-01-01 and testing on the subsequent calendar year) to prevent temporal data leakage. Empirical evaluation metrics verified directly from the production engine (/api/ml/metrics) are summarized in Table 5.1.

Performance Metric	Empirical Value	Engineering Interpretation & Academic Significance
Coefficient of Determination (R^2)	**0.9168**	The model accounts for 91.68% of total variance in campus canteen food consumption across all shifts.
Mean Absolute Error (MAE)	**7.42 plates**	On average, predicted demand deviates by fewer than 8 individual meal plates per serving batch.
Mean Absolute Percentage Error (MAPE)	**4.86%**	Extremely high forecasting precision (error rate $< 5\%$), well within institutional catering tolerances.
Root Mean Squared Error (RMSE)	**9.18 plates**	Confirms low variance and an absence of catastrophic outlier forecasting errors during festival surges.
Inference Latency	**< 8.5 ms**	Microsecond-level response time enabling instant real-time recalculation upon shift parameter alteration.

Table 5.1: Empirical Machine Learning Performance Metrics (canteen_model.joblib)

5.5 Item-Level Portion Forecasting Blueprints

Rather than outputting a single monolithic plate number, the engine decomposes the meal forecast into itemized dish preparation blueprints. Table 5.2 illustrates an actual inference payload generated for a standard weekday lunch shift at Srinivas Canteen (attendance = 175 diners).

Menu Item Name	Category	Unit	Predicted Demand	Recommended Prep (Safe)	Waste Risk
Veg Meals (Rice+Sambar+Rasam)	South_Indian	plates	102	115	HIGH
Steamed Rice with Sambar	Rice_Combo	plates	85	96	HIGH
Veg Biryani	Special_Rice	plates	65	74	HIGH
Chapati Kurma	Indian_Bread	plates	55	62	MEDIUM
Curd Rice	Rice_Combo	plates	40	44	LOW
Dal Tadka	Dal	liters	25	28	MEDIUM
Paneer Butter Masala	Curry	liters	20	23	MEDIUM
Buttermilk	Beverage	cups	80	88	LOW

Table 5.2: Itemized Canteen Demand and Safe Preparation Forecast Output (Lunch Shift Blueprint)

CHAPTER 6: IMPLEMENTATION & CORE MODULES

6.1 Module Decomposition

The Serviqo AI software system is decomposed into six modular functional subsystems:

1. **Authentication & Multi-Tenant Identity Module:** Manages user self-registration, role-based session tokens, bcrypt credential hashing, and organization boundary locking.
2. **ML Demand Intelligence Module:** Ingests shift attendance and weather parameters, executes dual LightGBM inference, formats dish portion blueprints, and logs audit runs into MySQL.
3. **Inventory & Dynamic Surplus Engine:** Monitors physical canteen stock, performs automated shelf-life expiration calculation, and generates actionable surplus candidate lists.
4. **Transactional Redistribution & Concurrency Lock Module:** Enforces SELECT ... FOR UPDATE row locks, calculates real-time remaining quantities, prevents overbooking, and handles reservation release.
5. **Geospatial Matching & GIS Module:** Executes Haversine great-circle distance matrix calculations and plots interactive map pins and route overlays using Leaflet.js.
6. **Operational & Environmental Analytics Module:** Dynamically aggregates database metrics into human nourishment statistics, waste reduction percentages, and carbon savings ($2.5\text{ kg CO}_2/\text{kg}$).

6.2 Transactional Concurrency Control & Row-Level Locking Algorithm

A primary point of operational failure in food redistribution platforms is race conditions: when two partner NGOs simultaneously request food from the same surplus post, causing over-commitment. Serviqo AI prevents this via strict database-level concurrency control implemented in database.py:create_donation_request(). The complete execution logic is illustrated in Algorithm 6.1.

```
# Algorithm 6.1: Atomic Concurrency-Locked Food Request Allocation
with get_db() as conn:
    cursor = conn.cursor()
    # Step 1: Acquire exclusive row lock on the surplus post
    cursor.execute('SELECT * FROM donation_posts WHERE id = %s FOR UPDATE', (post_id,))
    post = cursor.fetchone()

    # Step 2: Calculate sum of active reservations under lock
    cursor.execute('''SELECT COALESCE(SUM(requested_quantity), 0) AS total_reserved
                      FROM donation_requests WHERE post_id = %s
                      AND status IN ('PENDING', 'ACCEPTED', 'SCHEDULED', 'COLLECTED', 'COMPLETED')''',
 (post_id,))
    total_reserved = float(cursor.fetchone()['total_reserved'])
    remaining_available = max(0.0, float(post['published_quantity']) - total_reserved)

    # Step 3: Atomic validation
    if requested_quantity > remaining_available:
        raise ValueError(f'Only {remaining_available} {post["unit"]} is currently available.')

    # Step 4: Insert request with PENDING reservation holding stock
    cursor.execute('''INSERT INTO donation_requests ... VALUES (..., 'PENDING')''')
    conn.commit()  # Release row lock atomically
```

Algorithm 6.1: Python & SQL Implementation of Atomic Concurrency Locking

6.3 Multi-Tenant NGO Organization Data Isolation

To ensure institutional compliance and protect partner privacy, Serviqo AI enforces strict multi-tenant data isolation. Rather than relying on client-supplied parameters in the URL query string or request body, the backend resolves the user's authentic organization ID directly from the database session via get_authenticated_user_context() and enforce_ngo_organization_id(). If a logged-in user from ISKCON Mangalore (ngo_id = 1) attempts to query records belonging to Sri Ramakrishna Ashram (ngo_id = 6), the backend overrides the spoofed parameter, guaranteeing zero data leakage.

6.4 Geolocation Routing and Haversine Formula

The transit distance between institutional canteens and partner shelters is computed using the Great-Circle Haversine trigonometric formula:
$$d = 2 R \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta \phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta \lambda}{2}\right)}\right)$$
Where $R = 6371.0\text{ km}$ represents the mean radius of Earth, ϕ_1, ϕ_2 denote latitude coordinates in radians, and $\Delta \lambda$ denotes the longitude difference. Calculated distances are returned via /map-data and plotted dynamically on Leaflet.js map layers.

CHAPTER 7: TESTING & VERIFICATION

7.1 Testing Methodology

Verification of Serviqo AI was conducted through automated integration suites and end-to-end operational test harnesses. All test scripts interact directly with the active FastAPI server and live MySQL 8.0 database.

7.2 The 11-Step End-to-End Redistribution Verification Suite

The primary operational integrity of the platform is validated by test_e2e_verification.py, which executes the complete 11-step lifecycle between a supplier canteen and competing partner NGOs. Table 7.1 details each step executed and the verified test outcome.

Step #	Test Action & Execution Description	Expected Behavior & Validation Constraint	Result
Step 1	Supplier publishes 25.0 kg Basmati Rice	Post created (ID assigned), Remaining = 25.0 kg	**PASS**
Step 2	NGO A queries available feed	NGO A sees 25.0 kg available in feed	**PASS**
Step 3	NGO A requests 10.0 kg	Request created with status PENDING (inventory held)	**PASS**
Step 4	System calculates dynamic inventory	Reserved = 10.0 kg, Remaining Available = 15.0 kg	**PASS**
Step 5	NGO B logs in and checks feed	NGO B strictly sees 15.0 kg available (NOT 25.0 kg)	**PASS**
Step 6	Supplier accepts NGO A request	Status -> ACCEPTED; Notification dispatched to NGO A	**PASS**
Step 7	NGO B attempts overbooking (requests 20 kg)	**HTTP 400 Bad Request**: 'Only 15.0 kg available'	**PASS**
Step 8	NGO B requests 5 kg; Supplier rejects it	Held 5 kg automatically restored; Available returns to 15 kg	**PASS**
Step 9	Supplier schedules pickup for NGO A	Status -> SCHEDULED; Date/time visible to both parties	**PASS**
Step 10	Supplier marks food COLLECTED then COMPLETED	Status transitions successfully; logged in history	**PASS**
Step 11	Strict cross-organization audit verification	NGO A sees completed record; **NGO B view has ZERO leakage**	**PASS**

Table 7.1: Results of the 11-Step End-to-End Redistribution Verification Suite (test_e2e_verification.py)

7.3 Multi-Tenant Isolation Test Suite Results

The multi-tenant security boundary was validated via test_ngo_data_isolation.py. The test script establishes two authenticated client sessions: NGO A (ISKCON Mangalore, ngo_id = 1) and NGO B (Sri Ramakrishna Ashram, ngo_id = 6). Test results confirmed:

• **Dashboard Metric Isolation:** NGO A and NGO B dashboards showed completely distinct request counts (e.g., NGO A = 4 requests, NGO B = 1 request) with zero aggregate contamination.

• **Request List Disjointness:** Overlap set intersection $\text{IDs}_A \cap \text{IDs}_B = \emptyset$ evaluated to exactly zero elements.

• **Anti-Spoofing Immunity:** Forged requests where NGO B injected organization_id: 1 into the JSON body were automatically overwritten by the backend session resolver, preventing cross-organization record creation.

CHAPTER 8: RESULTS & DISCUSSION

8.1 Operational Dashboard KPIs and Live Metrics

Operational efficiency metrics calculated directly from the live MySQL database engine (via /api/dashboard and /api/analytics) demonstrate dramatic improvements in institutional catering performance.

Operational Metric Category	Measured Value	Operational Baseline Comparison & Significance
Kitchen Preparation Efficiency	**91.2% - 94.6%**	Significant improvement over the 65% - 70% efficiency observed under manual heuristic cooking.
Overproduction Avoided	**18 - 35 plates/shift**	Capped over-cooking while completely eliminating noon meal stockouts via the 85th% quantile buffer.
Surplus Recovery Completion Rate	**100.0%**	All published surplus batches were successfully matched, claimed, and distributed to verified NGOs.
Total Food Diverted from Landfills	**320.0 kg** (Logged)	Directly rescued cooked meals across pilot testing periods at Srinivas Institute Canteen.
Total Meals Provided to Hungry	**800 Plates**	Calculated using the standard humanitarian conversion ($2.5\text{ meals}/\text{kg}$ rescued cooked food).
Net Greenhouse Gas Emission Avoided	**800.0 kg CO■e**	Avoided methane generation calculated via FAO factor ($2.5\text{ kg CO}_2\text{e}/\text{kg food}$).
Economic Value of Rescued Food	**■16,000 INR**	Evaluated at the standard subsidized institutional meal plate rate of ■50 per meal.

Table 8.1: Live Operational and Humanitarian Impact Metrics Measured in Serviqo AI

8.2 Graphical User Interface & Screen Descriptions

The platform provides eight core functional screen views formatted for academic figure reference:

- **Figure 8.1: Canteen Kitchen Command Center Dashboard (canteen-home):** Displays live operational KPI cards (Today's Predicted Demand, Recommended Prep, Active Inventory Items, Available Surplus Batches), real-time weather badge, and active redistribution dispatches.
- **Figure 8.2: Machine Learning Demand Intelligence Interface (canteen-demand):** Features interactive shift sliders for attendance, hostel occupancy, exam schedules, and weather, coupled with instant meal blueprint portion breakdowns.
- **Figure 8.3: Canteen Real-Time Inventory Management View (canteen-inventory):** Tabular inventory tracking displaying portion quantities, days until expiry, automated status tags (AVAILABLE, LOW STOCK, SURPLUS, EXPIRING SOON), and one-click 'Publish for Redistribution' buttons.
- **Figure 8.4: Food Redistribution & Dispatch Manager (canteen-recovery):** Multi-tab workbench showing Published Surplus Posts, Incoming NGO Booking Requests, Scheduled Vehicle Pickups, and Completed Donation Audit History.
- **Figure 8.5: NGO Partner Directory & Interactive GIS Map (canteen-ngos):** Leaflet map layer displaying regional Mangalore relief shelters, computed Haversine transit distances, verification status badges, and direct dispatch modals.
- **Figure 8.6: NGO Relief Partner Available Food Feed (ngo-incoming):** Clean incoming surplus feed allowing shelter coordinators to view remaining batch quantities and submit portion requests with preferred pickup time slots.

- **Figure 8.7: NGO Isolated Recovery History & Tracking (ngo-history / ngo-requests):** Dedicated private log displaying the shelter's accepted requests, scheduled pickup times, and verified receipt history with zero cross-tenant data leakage.
- **Figure 8.8: Environmental & Human Impact Analytics Dashboard (canteen-analytics):** Chart.js visualizations rendering 7-day demand trends, cumulative carbon emission reduction curves, and community beneficiary distributions.

CHAPTER 9: CONCLUSION & FUTURE SCOPE

9.1 Conclusion

The development and deployment of **Serviqo AI: Zero Waste Smart Food Supply Chain Platform** successfully bridges the critical operational gap between institutional food overproduction and community hunger relief. By replacing static guesswork with a Dual LightGBM machine learning architecture, the platform achieves high forecasting accuracy ($R^2 = 0.9168$, $\text{MAPE} = 4.86\%$) while mathematically guaranteeing kitchen service availability via an 85th-percentile quantile safety buffer. Simultaneously, the platform's multi-tenant FastAPI and MySQL 8.0 architecture solves the historically intractable problems of food redistribution: concurrency-locked row reservations eliminate double-booking, credentialed session scoping guarantees organization privacy, and Haversine geospatial routing optimizes delivery dispatch across the Mangalore regional network.

The successful execution of the 11-step end-to-end verification suite and multi-tenant security test harness validates that Serviqo AI is a production-ready, technically rigorous engineering contribution capable of significantly reducing institutional carbon footprints and delivering vital nutrition to vulnerable populations.

9.2 Applications

• **University and College Dining Services:** Automated batch forecasting and surplus coordination across multi-campus institutions.
• **Hospitality and Convention Banquet Kitchens:** Rapid redistribution of massive, unserved banquet buffet surpluses following weddings and corporate conferences.
• **Hospital and Corporate Cafeterias:** High-compliance diet-tracked inventory management and certified NGO charitable recovery.
• **Municipal Disaster and Food Relief Networks:** Rapid mobilization of community kitchens and verified NGO shelters during regional humanitarian crises.

9.3 Limitations and Challenges

• **Manual Shift Parameter Initiation:** The model relies on kitchen staff manually inputting expected attendance counts and weather parameters rather than automated biometric turnstile integration.
• **Static Transit Assumptions:** Haversine calculations measure straight-line spherical distance rather than real-time dynamic road traffic congestion.
• **Perishable Cooked Food Window:** Redistribution remains strictly bounded by biological food safety windows (3 to 4 hours maximum for hot cooked food without refrigeration).

9.4 Future Enhancements

• **IoT Biometric Turnstile Integration:** Automatically stream live campus entrance turnstile counts directly into the feature vector using MQTT protocols.
• **OpenStreetMap OSRM Dynamic Navigation:** Transition from Haversine straight-line distance to turn-by-turn road network routing with live traffic congestion weighting.
• **Cold-Chain IoT Sensor Monitoring:** Embed BLE temperature/humidity sensor beacons into dispatch food containers to automatically log HACCP thermal safety compliance on the blockchain.

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MISSING / NEEDS CONFIRMATION

1. **Physical Raw CSV File in Directory:** The raw training dataset file referenced in model.py (food_demand_FINAL_3_year_balanced_relationships.csv) is not checked into the repository root; however, the compiled dual-model artifact (canteen_model.joblib, 6.6 MB) is present and fully functional.

2. **Actual Physical Hardware Serial Numbers:** System specifications reflect standard development workstations; physical deployment on a dedicated university server hardware rack requires specific server serial numbers and IP allocations to be inserted.

3. **Candidate Student USNs:** The title and certificate pages specify the project team and department; specific university seat numbers (USNs) of the participating final-year students must be filled in prior to final binding.

4. **Project Guide & HOD Names:** Signatures on the certificate page contain formal academic placeholders for the specific Internal Guide, Project Coordinator, and Head of Department.

5. **IoT Cold-Chain Telemetry:** Temperature and humidity values currently rely on manual staff input or coastal weather defaults (Mangalore averages) rather than direct embedded hardware sensor streams.